Exponential Decay for a Damped Wave Equation: A Minimal Test Fixture

Test Fixture[†]

[†]This is *not* a published paper. It exists only to exercise the well-posed pipeline end to end. The mathematics below is correct but elementary.

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Abstract

We consider the damped wave equation $u_{tt} + u_t - u_{xx} = 0$ on a bounded interval with homogeneous Dirichlet boundary conditions. We give a self-contained proof that the energy decays exponentially, with an explicit rate obtained from a multiplier identity. The note is written as a test fixture: it is short, it contains exactly one skipped step (so that a reader may practice filling it in), and its numbers can be checked by hand.

1 Introduction and main result

Let $L > 0$ and $\Omega = (0, L)$. Consider

$$u_{tt} + u_t - u_{xx} = 0, \qquad x \in (0, L),\ t > 0, \tag{1}$$

with Dirichlet boundary conditions

$$u(0, t) = u(L, t) = 0, \qquad t \geq 0, \tag{2}$$

and initial data

$$u(x, 0) = u_0(x), \qquad u_t(x, 0) = u_1(x). \tag{3}$$

The energy of a solution is

$$E(t) = \frac{1}{2} \int_0^L \left(u_t^2 + u_x^2 \right) dx. \tag{4}$$

Proposition 1 (Energy identity)**.** *Let u be a sufficiently smooth solution of (1)–(2). Then*

$$\frac{d}{dt} E(t) = -\int_0^L u_t^2 \, dx \leq 0. \tag{5}$$

Proof. Multiply (1) by u_t and integrate over $(0, L)$:

$$\int_0^L u_{tt} u_t \, dx + \int_0^L u_t^2 \, dx - \int_0^L u_{xx} u_t \, dx = 0.$$

The first term is $\frac{1}{2} \frac{d}{dt} \int u_t^2$, and integration by parts in the third term, together with (2), gives $-\int u_{xx} u_t = \int u_x u_{xt} = \frac{1}{2} \frac{d}{dt} \int u_x^2$. Collecting terms yields (5). □

Proposition 1 shows that E is nonincreasing, but it does not by itself give a rate. The main result of this note is the following.

Theorem 1 (Exponential decay)**.** *Let u solve (1)–(3) with $(u_0, u_1) \in H^2(\Omega) \cap H_0^1(\Omega) \times H_0^1(\Omega)$. Then there exist constants $C_1 \geq 1$ and $\mu > 0$, depending only on L, such that*

$$E(t) \leq C_1 \, e^{-\mu t} E(0), \qquad t \geq 0. \tag{6}$$

Remark 1. *The proof below is the classical multiplier argument. It is included in full because the constants are explicit and can be checked with a pencil; readers are invited to verify that the choice (13) below is legitimate.*

2 Proof of Theorem 1

Define the auxiliary functional

$$F(t) = \int_0^L u \, u_t \, dx + \frac{1}{2} \int_0^L u^2 \, dx. \tag{7}$$

Lemma 1. *Along solutions of (1)–(2),*

$$F'(t) = \int_0^L u_t^2 \, dx - \int_0^L u_x^2 \, dx. \tag{8}$$

Proof. Differentiating (7) and using (1) in the form $u_{tt} = u_{xx} - u_t$,

$$\begin{aligned} F'(t) &= \int_0^L \big(u_t^2 + u \, u_{tt}\big) dx + \int_0^L u \, u_t \, dx \\ &= \int_0^L u_t^2 \, dx + \int_0^L u \, (u_{xx} - u_t) \, dx + \int_0^L u \, u_t \, dx \\ &= \int_0^L u_t^2 \, dx - \int_0^L u_x^2 \, dx, \end{aligned}$$

where the boundary term $\big[u \, u_x\big]_0^L$ vanishes by (2) and the two terms involving $u \, u_t$ cancel. □

Now set, for $\varepsilon > 0$ to be chosen,

$$E_\varepsilon(t) = E(t) + \varepsilon F(t). \tag{9}$$

Combining (5) and (8),

$$E_\varepsilon'(t) = -(1 - \varepsilon) \int_0^L u_t^2 \, dx - \varepsilon \int_0^L u_x^2 \, dx. \tag{10}$$

It remains to compare E_ε with E. By the Cauchy–Schwarz inequality and the Poincaré inequality one obtains

$$|F(t)| \leq C_0 E(t), \qquad C_0 = C_0(L), \tag{11}$$

so that for ε small enough,

$$(1 - \varepsilon C_0) E(t) \leq E_\varepsilon(t) \leq (1 + \varepsilon C_0) E(t). \tag{12}$$

Choose

$$0 < \varepsilon < \frac{1}{2C_0}, \qquad \varepsilon < \frac{1}{2}. \tag{13}$$

Then (10) gives, for some $c = c(\varepsilon) > 0$, $E_\varepsilon' \leq -cE$, and by (12), $E_\varepsilon' \leq -\frac{c}{1+\varepsilon C_0} E_\varepsilon$. Gronwall's inequality yields

$$E_\varepsilon(t) \leq e^{-\mu t} E_\varepsilon(0), \qquad \mu = \frac{c}{1 + \varepsilon C_0},$$

and applying (12) once more gives (6) with $C_1 = (1 + \varepsilon C_0)/(1 - \varepsilon C_0)$.

Remark 2. *The estimate (11) is the only place where the boundary conditions enter the argument beyond (5); without (2) the Poincaré inequality is unavailable and F cannot be dominated by E.*

3 Conclusion

The proof uses three ingredients: an energy identity (Proposition 1), a multiplier identity producing a functional whose derivative has the opposite sign (Lemma 1), and an equivalence estimate that turns the modified energy into the original one. This is the same scheme used for variable-coefficient and higher-dimensional versions of (1).

References

[1] V. Komornik, *Exact Controllability and Stabilization: The Multiplier Method*, Masson/Wiley, 1994.

[2] J.-L. Lions, *Contrôlabilité exacte, perturbations et stabilisation de systèmes distribués*, Masson, 1988.

[3] E. Zuazua, Exponential decay for the semilinear wave equation with locally distributed damping, *Comm. PDE* 15 (1990), 205–235.